Google Cloud

Importance of automation in the data life cycle

Business demands are always changing and, as a data professional, being able to adapt and respond is invaluable.

Sometimes data professionals will wish they can multitask or be in two places at once.

This actually applies to many jobs!

For example, consider a farmer.

When the farm is small, a farmer might water crops by herself every morning.

As the farm grows, the farmer no longer has time to water the crops as she once did.

She wants to figure out how to water a huge field's worth of crops efficiently.

How can she possibly achieve this?

She invests in a timing system that waters her crops at the same time every day for a preset interval.

This means that all of her crops are watered simultaneously and she harvests a record number of crops.

In data management, you'll sometimes encounter similar situations!

You might need to ingest data on a set schedule or you may need to automatically cap the amount of data ingested.

You might be asked to be more efficient with a set amount of resources.

There are many ways you could be asked to scale or streamline data processing.

In this video, we'll explore how data management systems can be implemented and automated to respond to ever-changing business needs.

Automation is the use of software, scripting, or machine learning to perform data analysis processes without human work.

For example, consider a business that has over two hundred employees to pay twice per month.

Someone could verify the hours worked for all employees.

Then, another person could calculate the amount of each paycheck.

And a third person could write and sign the checks.

Most businesses don't operate this way!

Instead, the payroll process is usually automated.

Verification of hours worked and calculation of pay is all done automatically by a computer program.

In some cases, payment is automatically transferred to each employee's bank account by an automated process.

Automation saves time and resources —and ensures that people are paid on time!

One of the key benefits of automation is that it reduces mistakes during data collection through automatic checks for errors or incomplete data.

This helps ensure your dataset is complete and accurate.

In situations where compliance is key, automation ensures uniformity and accuracy important for compliance.

Automation scales up workloads —ingesting or processing more data at once— without losing key information.

Finally, automation allows for efficiency and process improvement, which can help to preserve resources —both human and technical— and controls costs.

Having a clear process for moving data through a cycle makes it easier to control your outcomes.

With automation added at key areas, you'll be able to move data through its lifecycle quickly and efficiently.

Now that you've automated repetitive tasks, this frees up your time to focus on the meaningful work of data analysis!